API GATEWAY

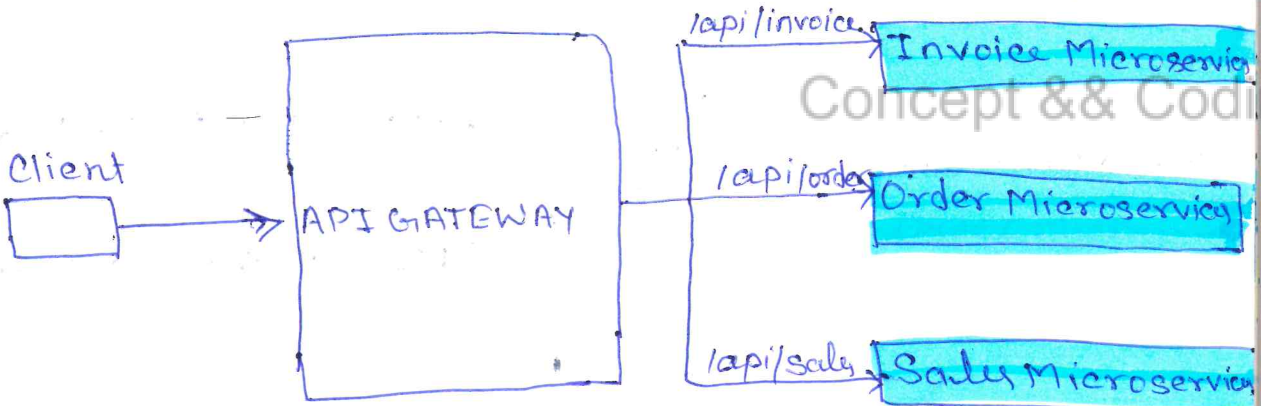
* If API GATEWay a Single entry point, how does it handle millions of requests per second?

also

* How Api Gatway is different from Load Balancer? Does it comes before or after the load balancer?

What is Api Gateway?

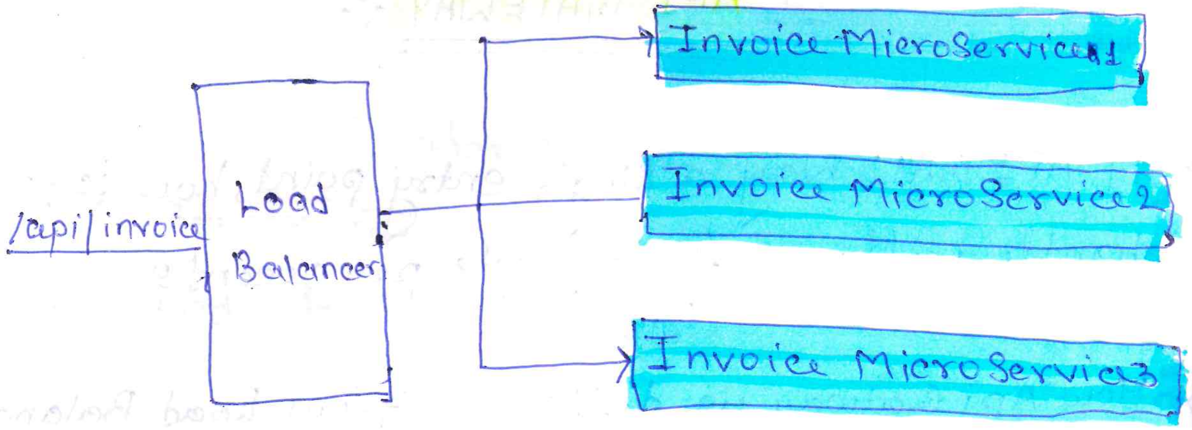
- It accepts the client API request and route them to correct backend service based on API endpoint.



Isn't this what load balancer also do?

No, generally Load Balancer simply distribute the traffic to the multiple instances of a microservices based on various factor like health, trafic load etc.

But usually they don't have capability to understand an API and then take decision, where to route it.

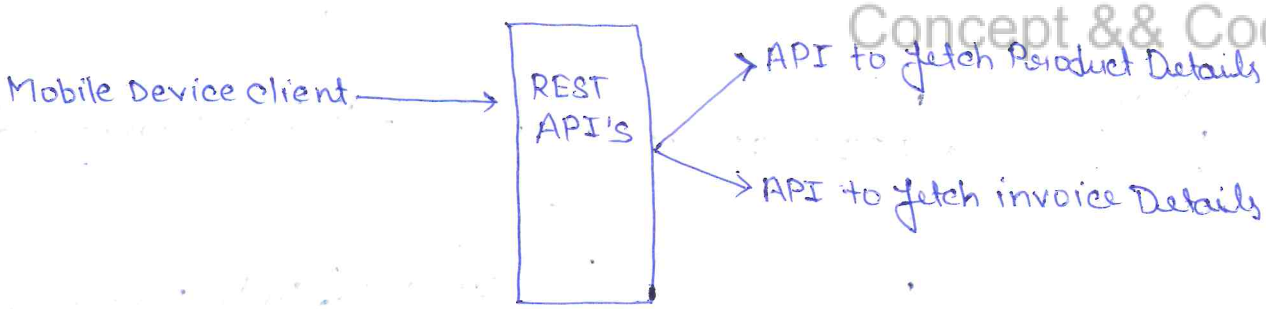


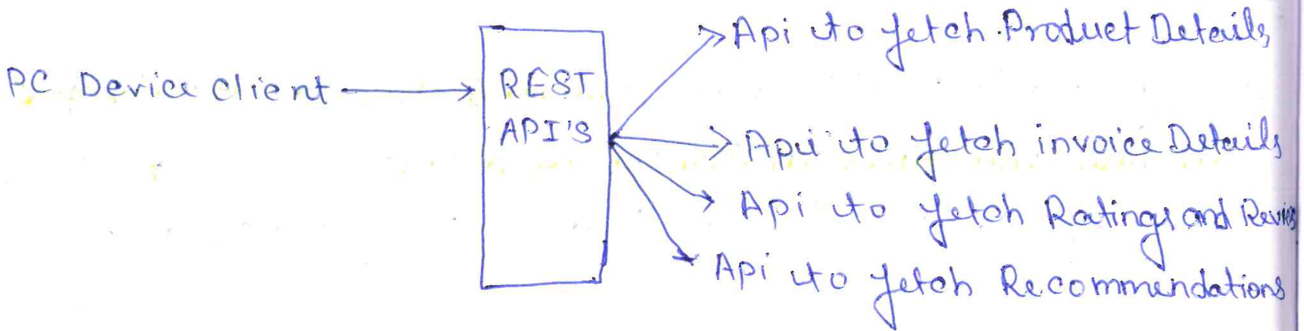
Other Capabilities of API Gateway :-

1:- API Composition

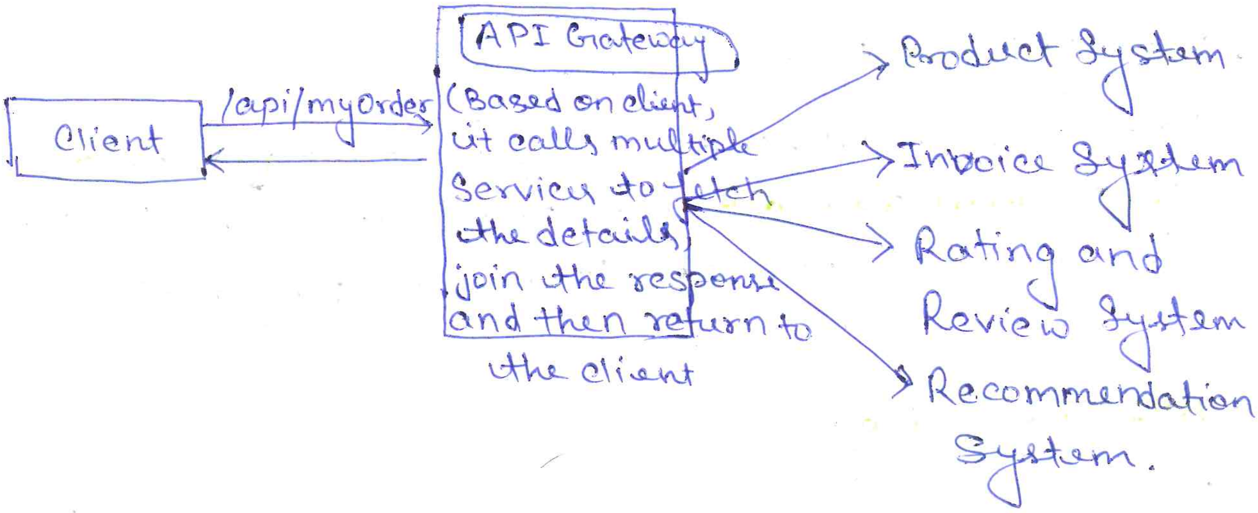
very frequently use in Netflix

Problem : Let say, we are visting "My Orders" page.



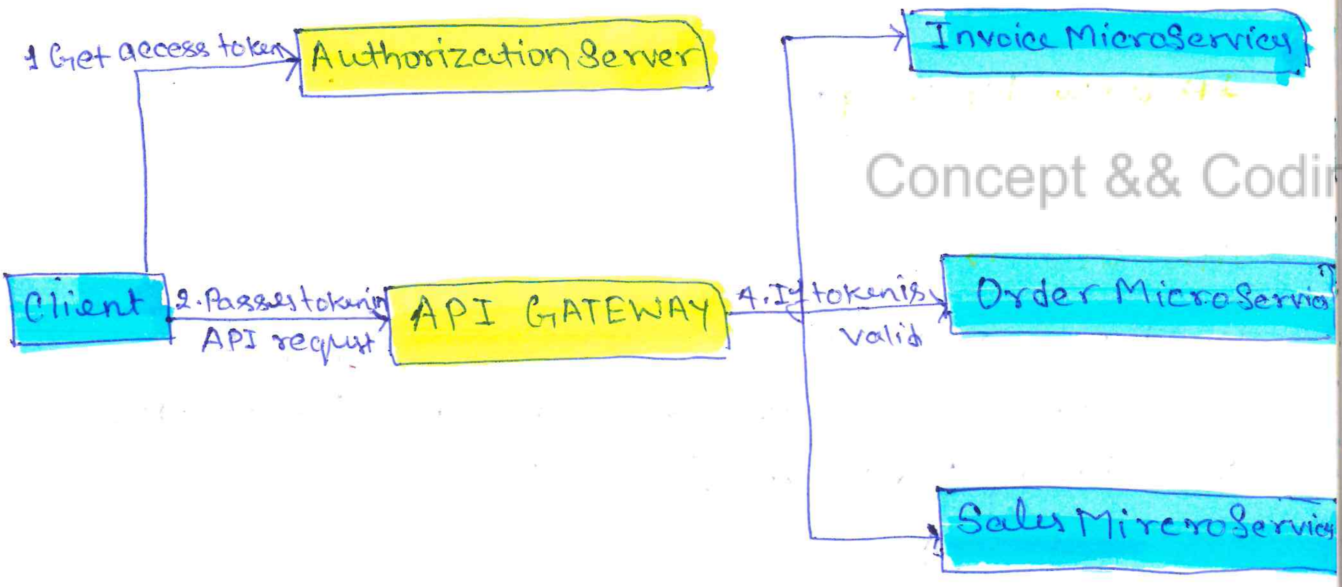


API Gateway, can make life of client easier through Api Composition functionality.

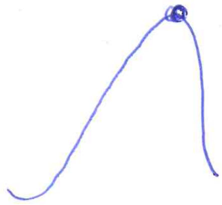


2. Authentication

Like OAUTH 2.0 flow



3:- Rate Limiting



Brust limit = 500

We can set rules like :

- **Manage Burst Limit :**

  Use to limit the Burst traffic, means max no of concurrent request that Api Gateway can handle before it return 429 (Too many Requests)

- **API Throttling :-**

  Limiting the numbers of requests from an individual or an application by temp blocking the request, once they crossed the allowed request rate.

- **Ip base blocking :-**

- **API Queues :-**

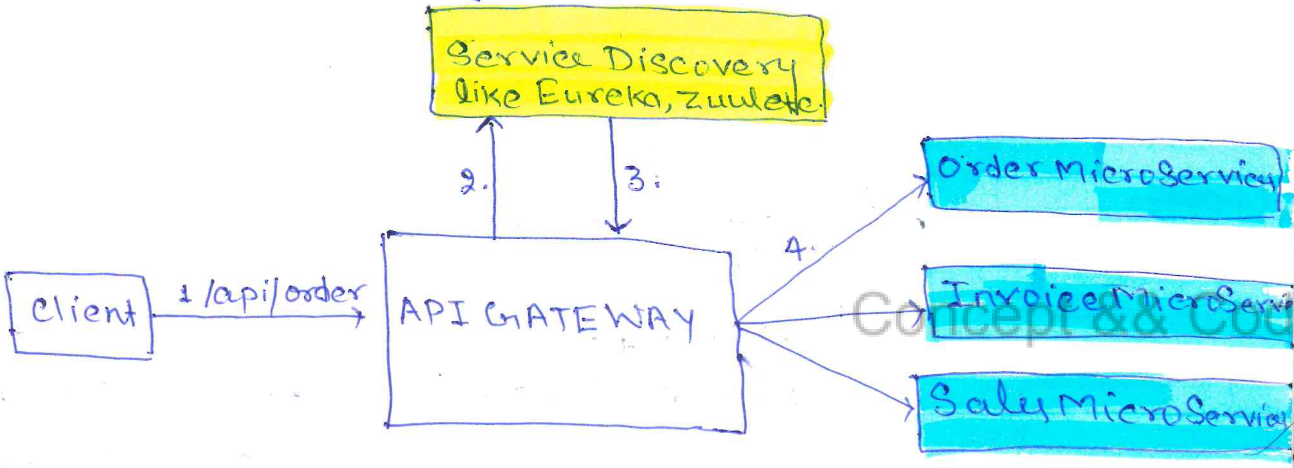
  Hold requests to an API, which can not be processed immediately. It helps to handle the Thundering herd issue.

4. Service Discovery

As microservices can scale up and down, its necessary to know the location (IP address and Port). Service Discovery keep tracks of those.

1 Approach 1 :- Each MicroServices register or de-register themselves.

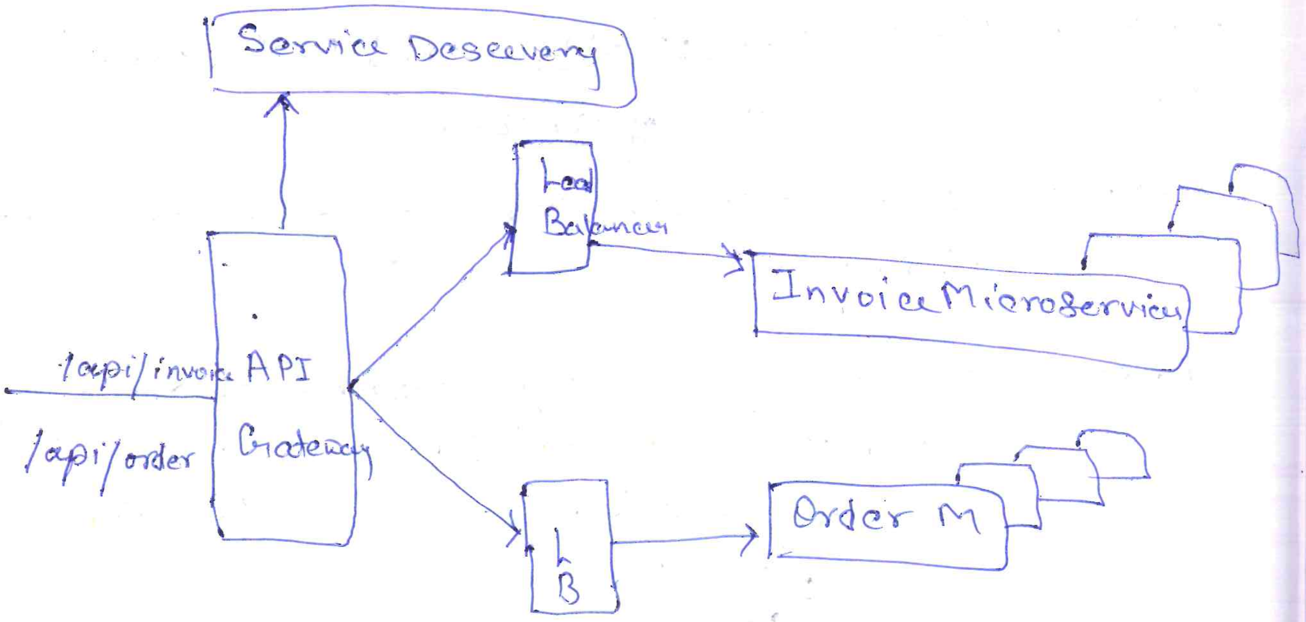
Approach 2 :- Service Discovery, keeps health check of all the registered microservices and keep only active microservices location.



And many other capabilities like :

- Request - Response transformation
- Response caching
- Logging

Q] API GATEWAY a single entry point, how does handle millions of requests per second?



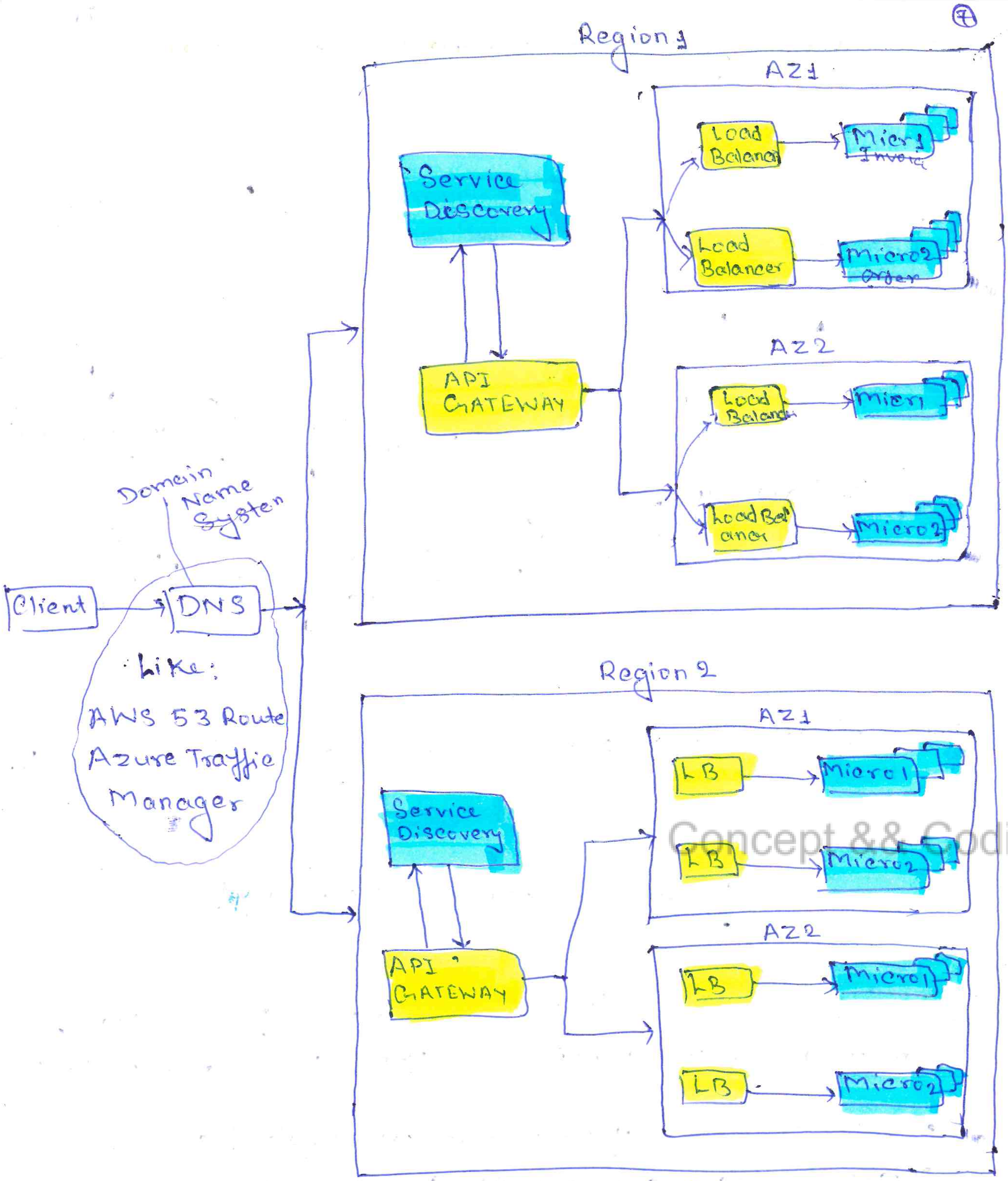
Region 1 → Mumbai

~~Ability~~ AZ 1 →
availability zone → area 1 = Bandra
AZ 2 → = area 2 = Santacruz
AZ 3 → area 3 = Andheri

dedicated data centre

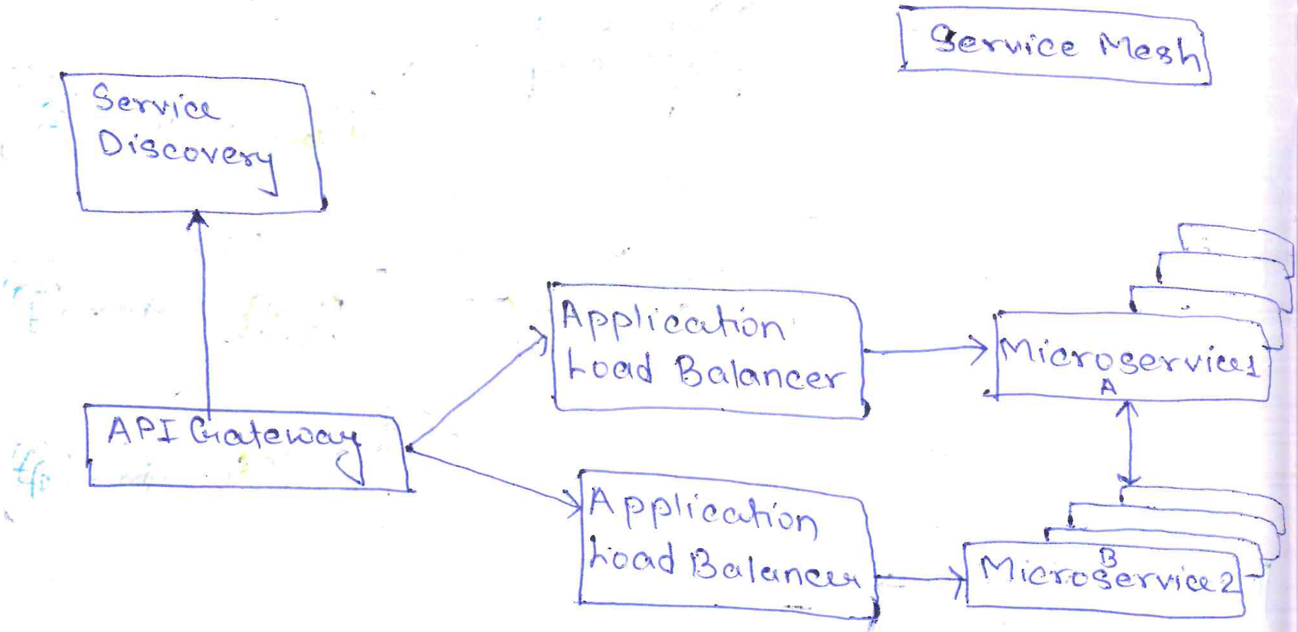
Region 2 - Banglore

AZ1 → (
AZ2 → (-)



* DNS can routes the traffic based on region

How 2 Micro System communicate with each other

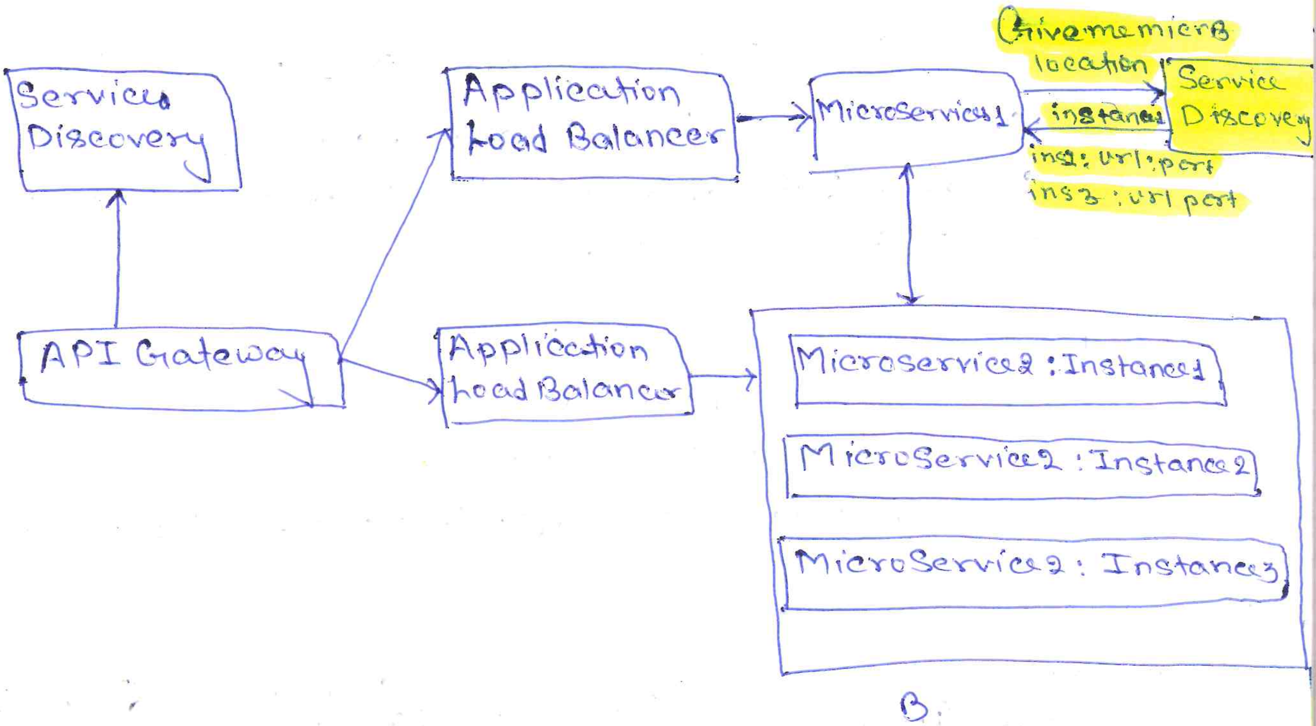


Need of MicroServices A to Communicate with MicroserviceB.

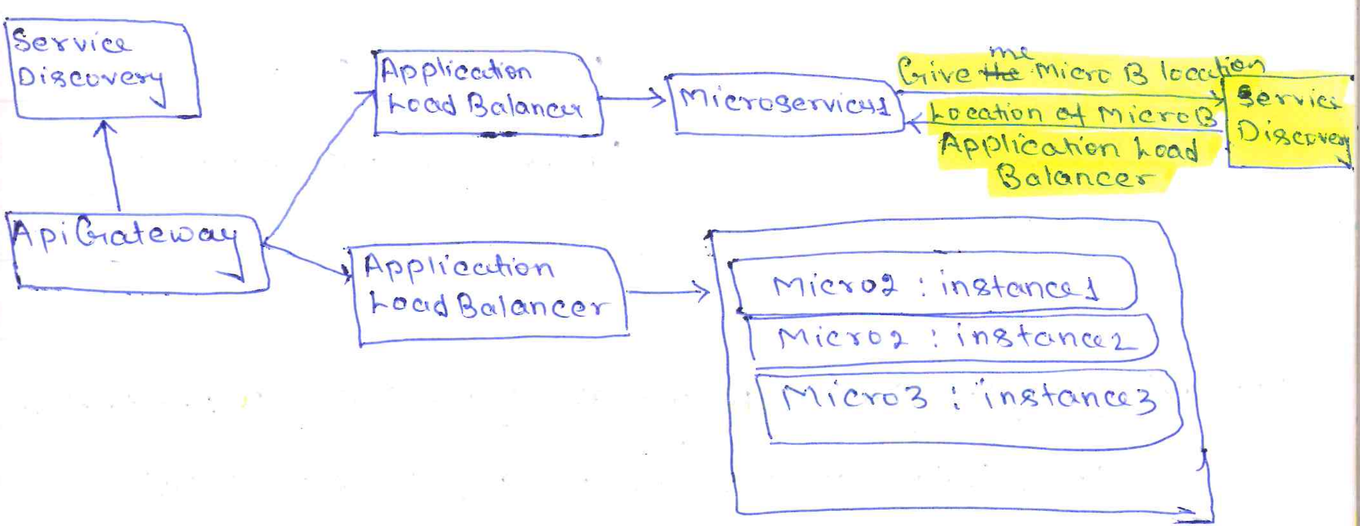
1. Service Discovery Capability

- MicroServices A wants to communicate with MicroService B. Then it need its ~~loc~~ location (IP Address and port no).
- Service Discovery functionality is required so that Microservices A can get the location of Microservices B different instances.

Scenario 1:- Service Discovery returns all the MicroserviceB instances location.



Scenario 2:→ Service discovery returned the address of Microservice B Application Load balancer.



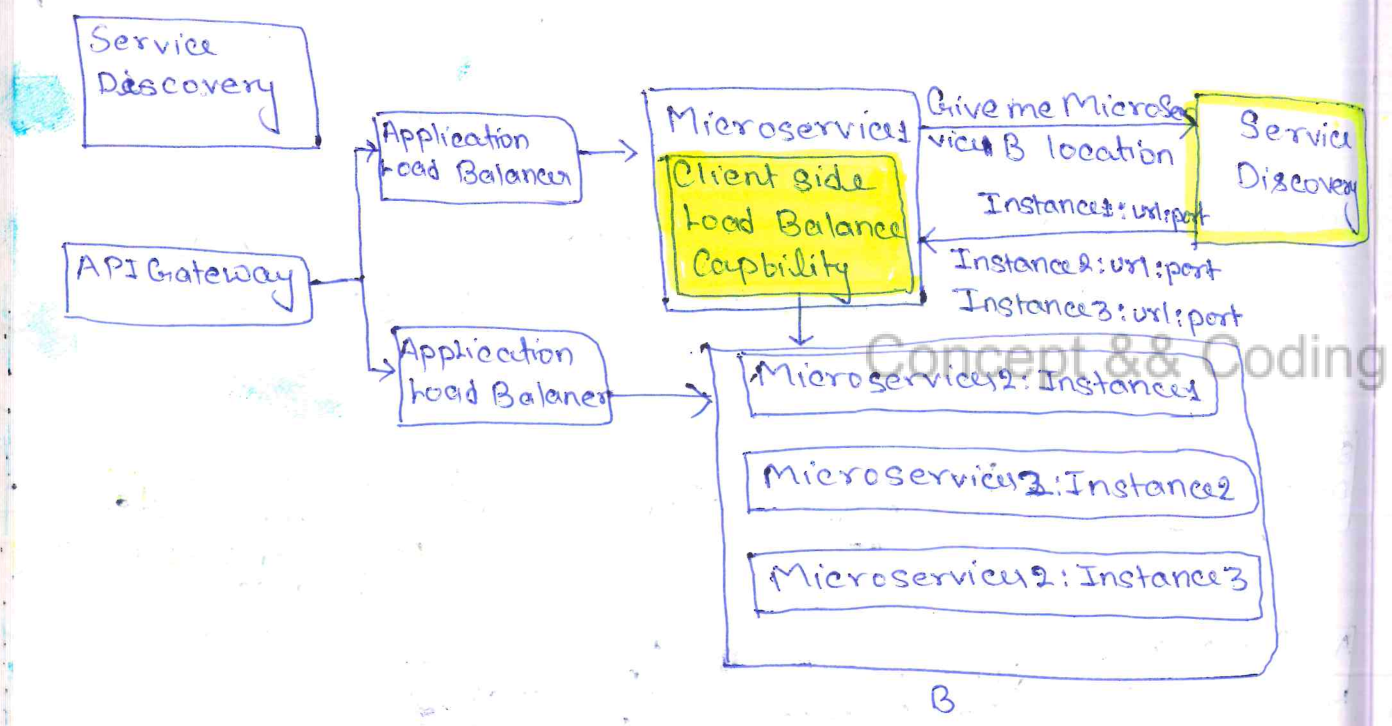
2. Load Balancing Capability:

Scenario 1: MicroServiceA got the MicroServiceB instances location and now want Load Balancing capability to equally distributed the distributed the traffic to MicroServicesB.

MicroServicesA → MB

application.properties

microserviceB.url = {instance-url}:{port-no}

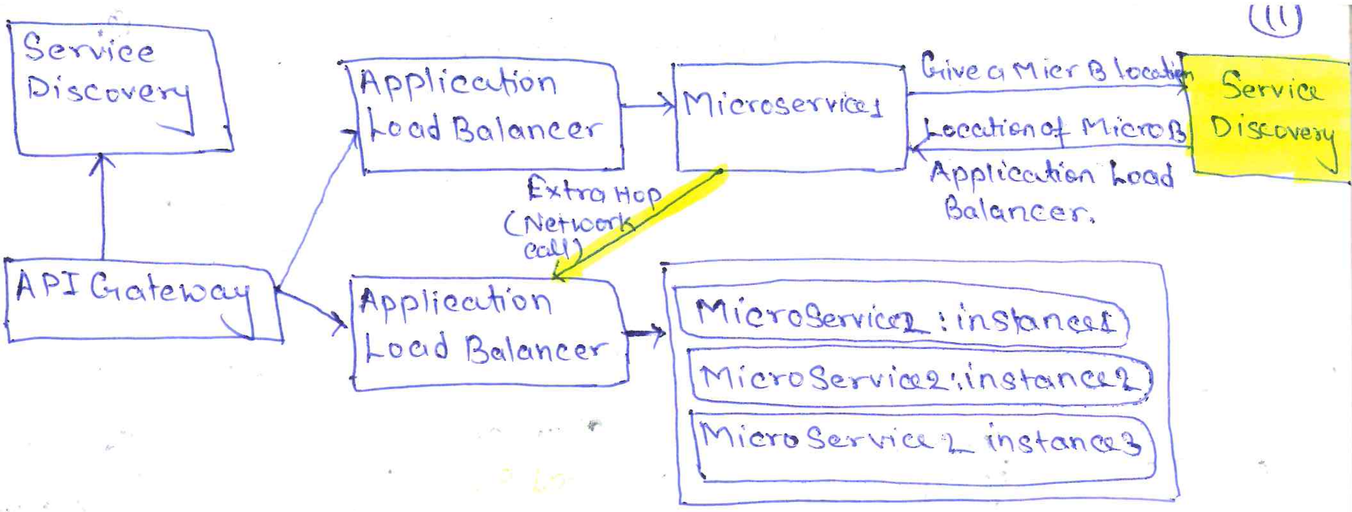


Scenario 2: MicroServicesA got the MicroServicesB Application load balancer address and now need to invoke that.

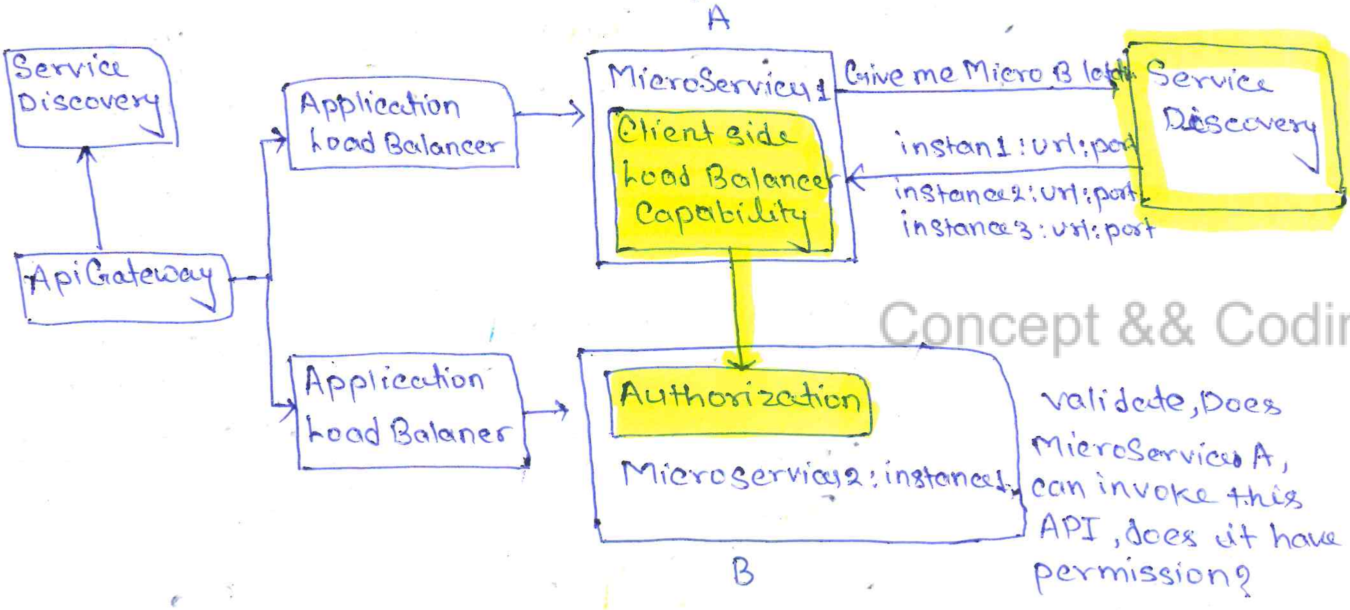
- Extra Network call is required, which might cause latency issue.

application.properties

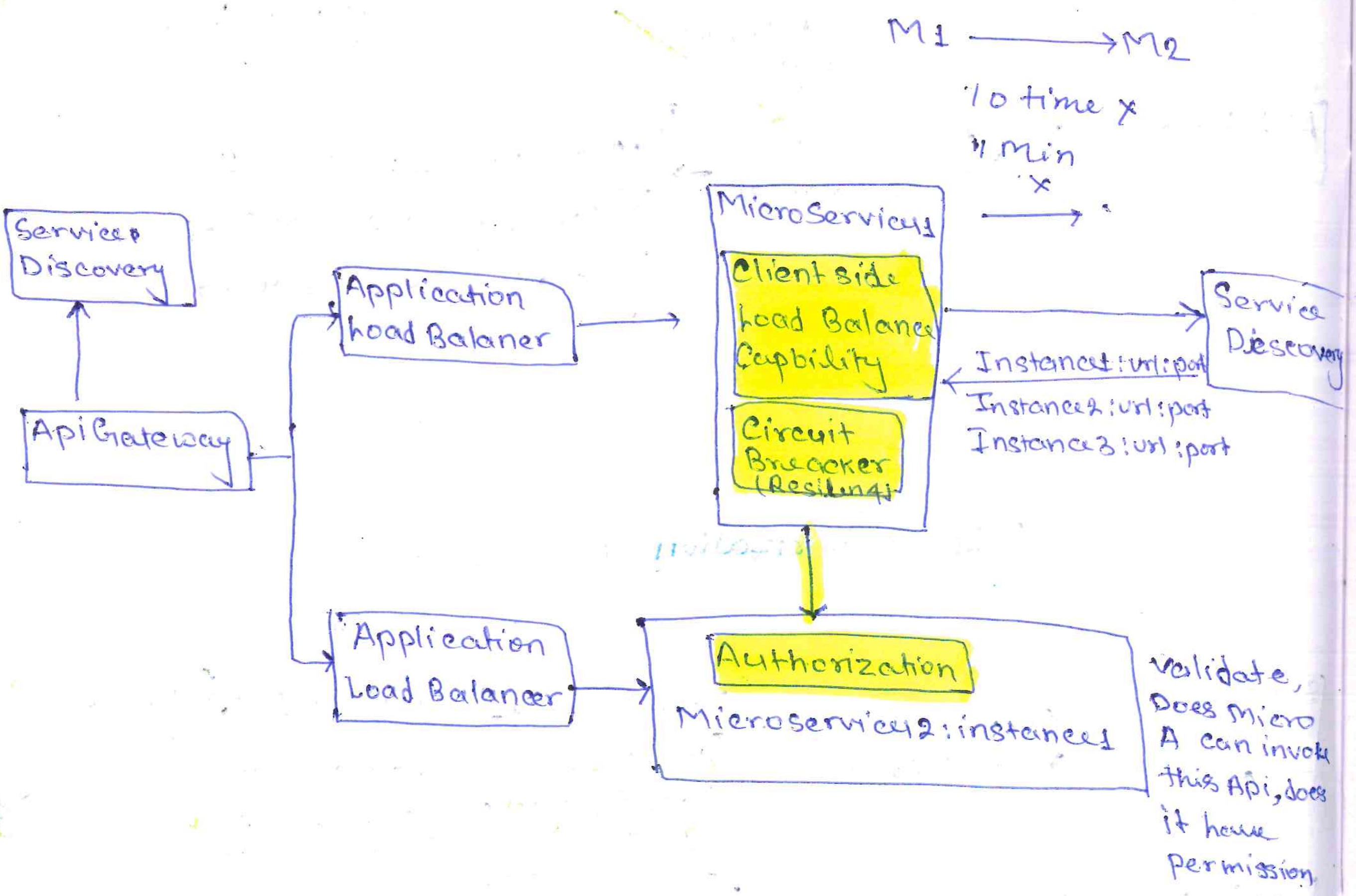
microservicesB.url = {application-load-balancer-url}: {port-no}



3. Authorization / Authentication

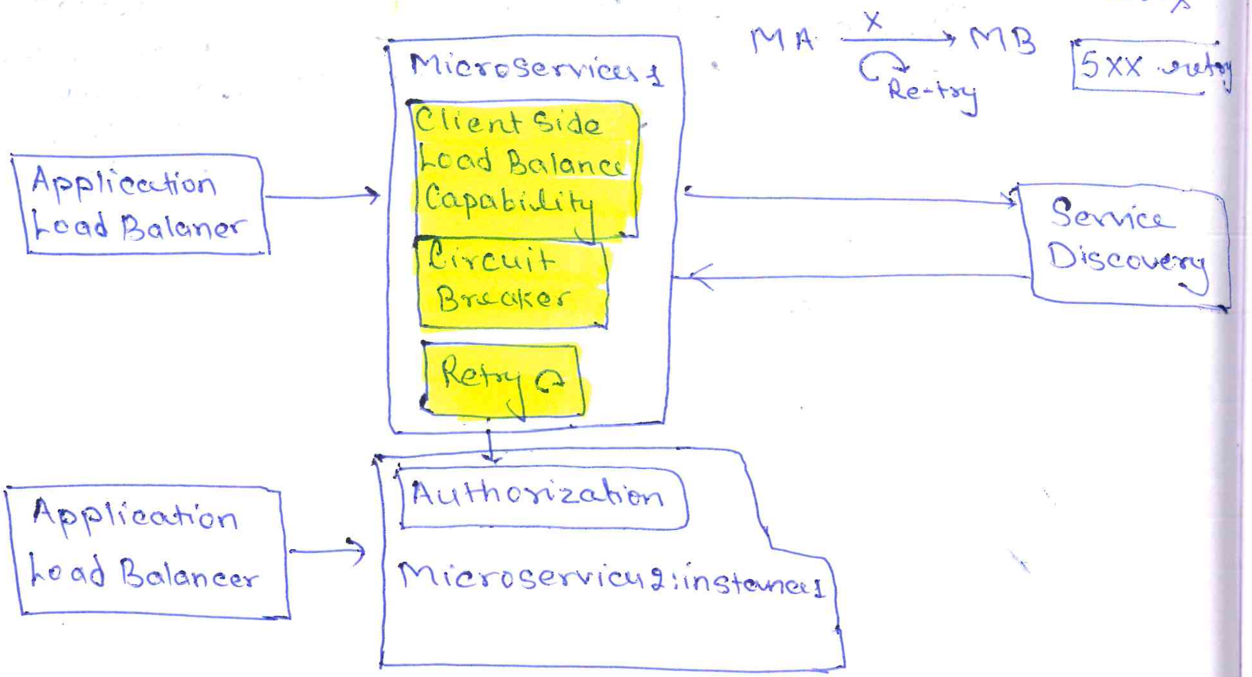


4. Circuit Breaker Capability



5. Re-try Capability

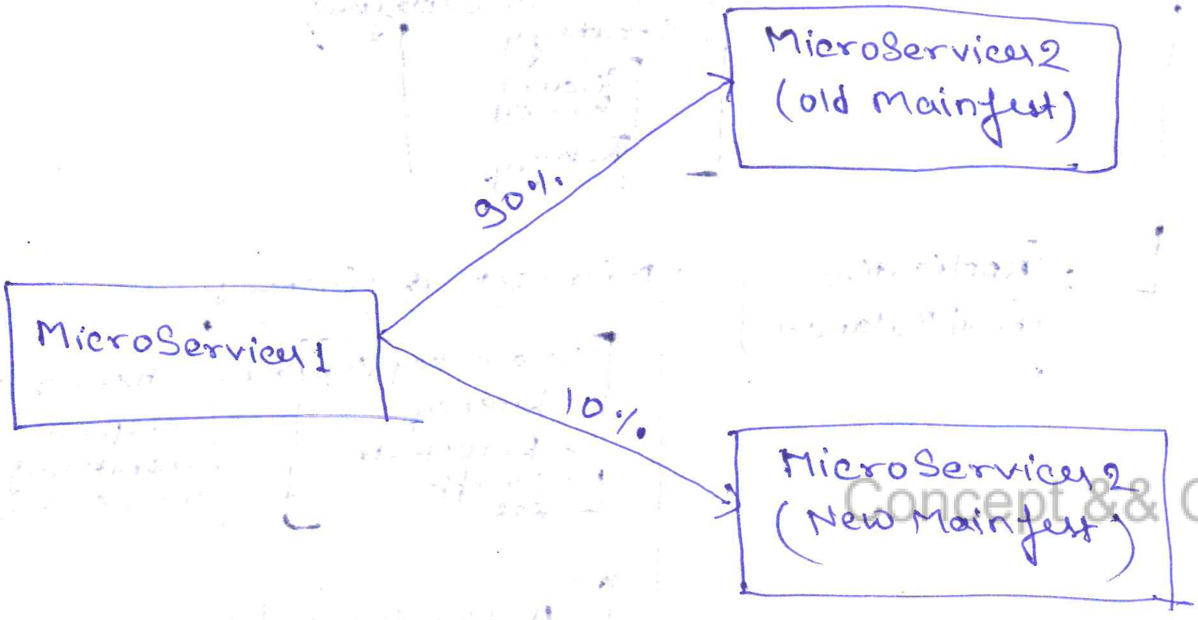
4xx } validation error X



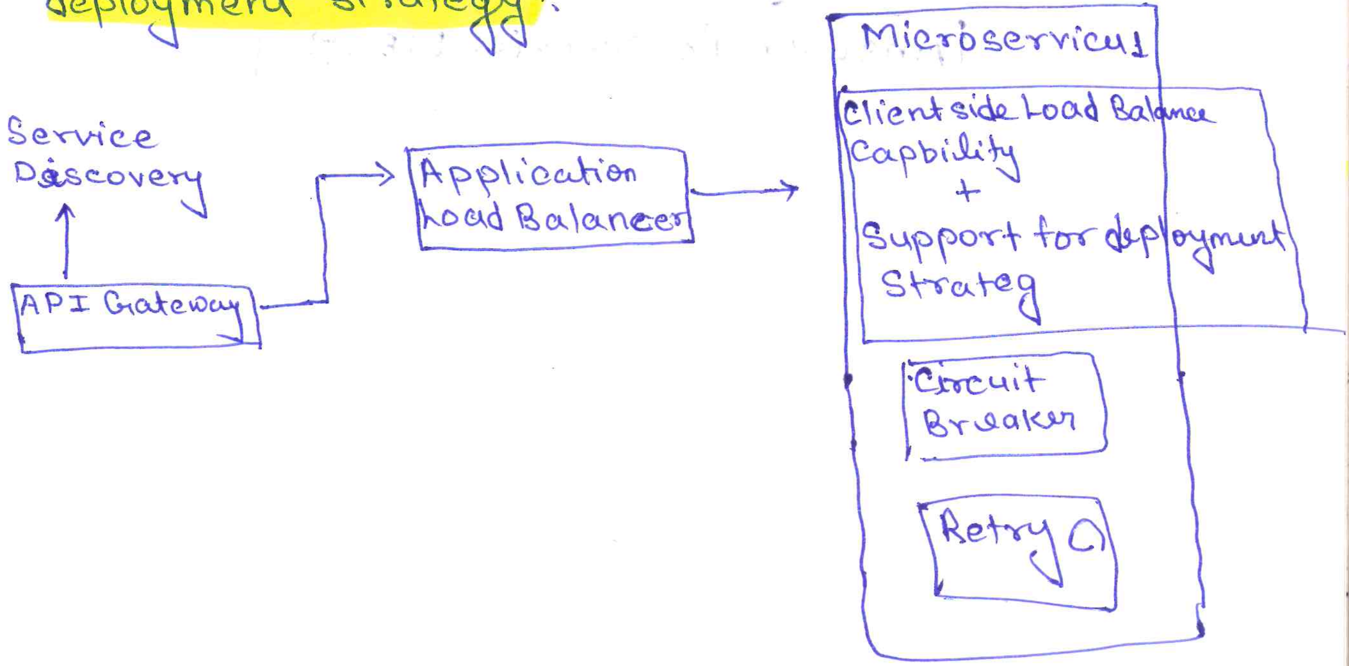
6. Deployment Strategy:

There are different deployment strategy like Canary, Red-Black, Blue-Green etc.

For example: In Canary, I want 90% traffic on Old and 10% on new initially, and gradually increase the number of ~~ra ce~~ new manifest.

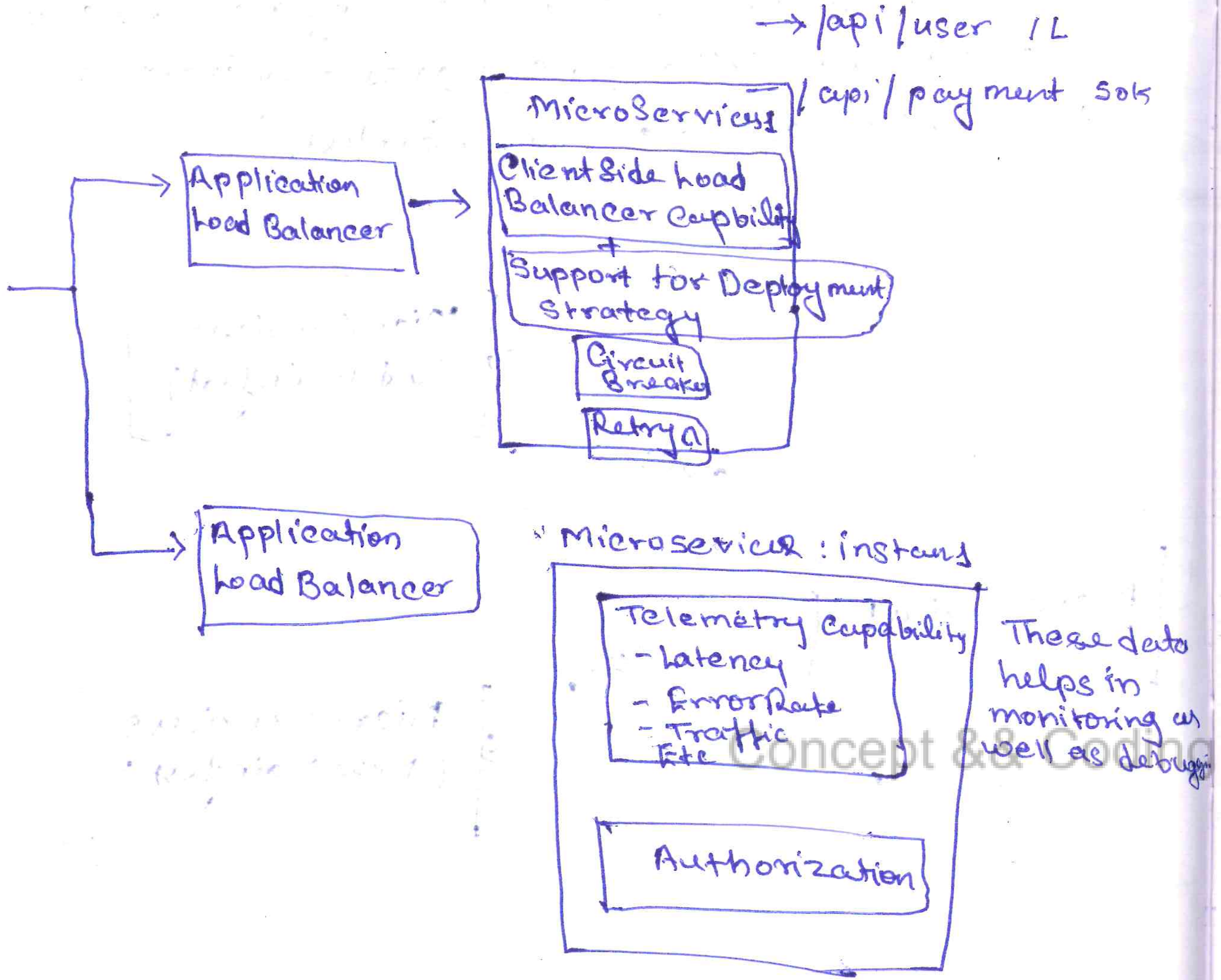


* Generally load blancer helps to control this, means "client side load balancer also has handle this different deployment strategy".



7. Telemetry Capability

- It means, we need to collect data which will help in analysis and monitoring of a Component.



Any better Solution?

Answer is: Service MESH